

Rayan Butt

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Studied **Computer Science at the University of York** specialising in **AI, computer vision, and full-stack development**. Experienced in **Deployment, Testing and Databases** with projects in **Go web scraping and analysis**, and **web application development**. Strong **problem-solving skills**, **adaptable in fast-paced environments**, and **eager to contribute to innovative projects and learn new skills**. Always ready to **take charge** and tackle problems with **unique and tailored approaches**.

Technical Skills

- **Languages:** Python, Go, JavaScript, TypeScript, Java, SQL, HTML, CSS, C++
- **Frameworks & Libraries:** Pytorch, Flask, Node.js, React.js
- **Tools & Platforms:** Postgres, GitHub Actions (CI/CD), Linux, Windows, App development, Microsoft Copilot Studios.
- **Specialist Knowledge:** Front-end and back-end development, AI & Machine Learning, Computer Vision, Data Structures & Algorithms, User-Centred Design, Network Administration.
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Relevant Projects

Budgeting Flask Web App (Postgres) - <https://github.com/RayRB21/Budgeter>

Developed, deployed and maintained a personal budgeting web application which uses a linked **database**, enabling users to record income and expenses, set savings goals, and view spending trends. The project incorporates **Python, HTML/CSS, and Postgres for backend, frontend and data storage**, demonstrating skills in **full-stack development, API design, and database integration**. It also incorporates a **CI/CD Pipeline with GitHub Actions** to **automate testing** and improve deployment reliability.

Trading Card Price Tracker (Golang API) - <https://github.com/RayRB21/tcg-price-tracker>

Built a **Go CLI tool** that **reverse engineered PriceCharting's internal paginated JSON endpoint** — discovered via Chrome DevTools network analysis — to **scrape live prices across 700+ Riftbound trading cards**, persisting time-series snapshots to a **SQLite database** using versioned migrations. Implemented a pure **Go analysis engine** to **compute price movement** across stored snapshots, surfacing spike detection with a configurable threshold, top movers ranked by absolute percentage change, and per-card price history with moving averages. Designed a **cobra-powered CLI** with subcommands (scrape, movers, spikes, history) **enabling database queries** independent of the scraper, with idiomatic Go package structure across internal/scraper, internal/storage, and internal/analysis packages.

Dissertation - Face fitting with robust image feature descriptors:

The project focused on evaluating a technique for analyzing images, especially faces, using an **Active Appearance Model (AAM)**, which combines shape and appearance information. This model was enhanced with robust feature descriptors and tested on images collected in **uncontrolled “in-the-wild” environments**. The goal was to **accurately capture the shape of objects by fitting landmark points**, a key step in many computer vision tasks such as facial recognition and human-computer interaction.

Portfolio Website (React - Typescript) - <https://github.com/RayRB21/Portfolio-Website>

Built and deployed a **responsive personal portfolio** website using **React (TypeScript)** and **Bootstrap**, featuring **animated UI components**, **scroll-triggered effects**, and **reusable architecture**, hosted on Netlify with continuous deployment via GitHub.

Education

2021 – 2024 University of York

Diploma of Higher Education (2:1)

Relevant Modules:

- Data 2: Data Analysis and Management (**DATA**)
- Human-Computer Interaction 2: User Experience (**USER EXPERIENCE**)
- Intelligent Systems 1: Introduction to Artificial Intelligence (**AI**)
- Computer Vision & Graphics (**AI**)
- Evolutionary & Adaptive Computing (**AI**)

2013 – 2020 Parmiters School Watford

Employment History

Jan 2025 - Dec 2025

Network Technician

VRXTRA

- Managed and supported **network operations for a 10-station setup**, including PCs and VR headsets
- Provided on-site **technical support, troubleshooting hardware, software, and network issues**
- **Responded to unexpected outages and performance issues** to maintain service availability

Jun 2023 - Current

Tutor

Self-employed/ Parmiters School

- Delivered coding lessons to **groups and 1-to-1 students**, supporting GCSE and A-level Computer Science
- Designed lesson plans and **adapted explanations** to suit different learning styles
- Helped students **achieve top grades** through **structured problem-solving** and exam preparation

Jun 2020- Sept 2020

Software Engineering Intern

LMKR Software Company

- Collaborated within a small development team to deliver **internal software solutions**
- **Gathered user requirements** and **iterated on features based on stakeholder feedback**
- **Developed and deployed Python and JavaScript algorithms** that improved task efficiency
- Recognised as a **top-performing intern** for technical contribution and teamwork